

Skyrmions and Hopfions in 3D Frustrated Magnets

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July 8, 2022

Outline

- Naya, D.S., Shifman, Wang (2021) arXiv: 2111.06385
- 3D topological defects in a lattice spin model
- Introduction
 - ▶ What is a 2D Skyrmion? (baby Skyrmion)
 - ▶ What is a 3D Skyrmion? (baryon)
 - ▶ A bit of geometry involving $S^3 \rightarrow S^2$
 - ▶ Squashed sphere model
- Lattice spin models and baryons
- Extension of baby Skyrmions to 3D loops of string

Heisenberg model and 2D Skyrmions

- Classical ferromagnetic Heisenberg model

$$H = -J \sum_{\langle x,y \rangle} S(x) \cdot S(y) \quad \sim \quad \frac{J}{2} \int d^2x (\partial S(x))^2$$

- Target space is S^2 . $\pi_2(S^2) = \mathbb{Z}$, so topological defects in 2D
- Derrick's theorem: Must be additional higher derivative terms
- Frustrated interactions rather than Dzyaloshinskii-Moriya
- “baby Skyrmions” or “magnetic Skyrmions”

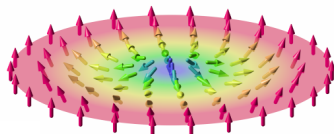


Image: Karin Everschor-Sitte and Matthias Sitte (Wikimedia Commons)

Principal chiral model and 3D Skyrmions

- Low energy effective theory of QCD in terms of $U \in SU(N_f = 2)$

$$\mathcal{L}_{PCM} = \frac{f_\pi^2}{4} \text{Tr} (\partial_\mu U^{-1} \partial_\mu U) .$$

- A column of U is a 2-component complex unit vector $z \in S^3$

$$U = \begin{pmatrix} \bar{z}^1 & z^0 \\ -\bar{z}^0 & z^1 \end{pmatrix}, \quad \mathcal{L}_{PCM} = \frac{f_\pi^2}{2} \partial \bar{z} \cdot \partial z.$$

- Again topological defects since $\pi_3(S^3) = \mathbb{Z}$. Stable if Skyrme term
- $Q = 1$ ‘hedgehog’ solution. $Q > 1$ less symmetric rational map solutions

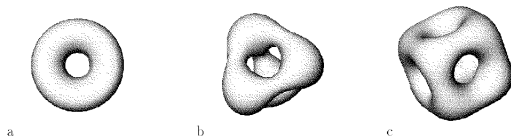


Image: Houghton, Manton, Sutcliffe (1997)

Mapping S^3 to S^2

- Can relate the S^3 and S^2 sigma models more directly
- If we ignore the phase of z , this is CP^1 . Bloch sphere
- More concretely, $S^i = -\bar{z}\sigma^i z$. Does not depend on phase
- If we use this change of variables,

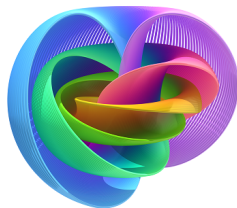
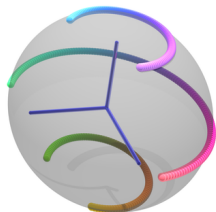
$$\frac{1}{4} (\partial S)^2 = \partial \bar{z} \cdot \partial z - |\bar{z} \cdot \partial z|^2$$

- Additional term subtracts out any change of z in phase direction
- Could imagine only partially subtracting out this change

Squashed sphere

- Hopf fibration visualization
- Generically $SU(2) \times U(1)$ global symmetry
- Can adjust length of fibers
- Zero length: $U(1)$ is gauged
- Special length: $SU(2) \times SU(2)$
- Intermediate length parametrized by κ

$$\partial \bar{z} \cdot \partial z - \kappa |\bar{z} \cdot \partial z|^2$$



Squashed sphere model in high-energy physics

- In 2D, CP^{N-1} model toy model for 4D gauge theories
- Couple CP^{N-1} model to extra degree of freedom
 - ▶ Stueckelberg field
 - ▶ Axion field (T-duality)
 - ▶ Massless fermion (bosonization)
- This can restore integrability
- $N = 2$: η deformation (Delduc, Magro, Vicedo '14)
- $N > 2$: AdS_4/CFT_3 duality: Bykov '10, Basso, Rej '12, '13
- In thesis: Cancellation of renormalons along integrable trajectory

Squashed sphere model as an effective field theory

- EFT for spin systems with a non-collinear ground state
- 2+1d: Dombre, Read '88; Chubukov, Sachdev, Senthil '94
- 3d: Can show by an ϵ -expansion there is Wilson-Fisher fixed point
- Azaria, Lecheminant, Mouhanna '95: Large N vs weak coupling
- In thesis: discrepancy just due to scheme dependence
- Large N agrees with wave function methods (D.S. '21)
- Weak coupling agrees with WZW CFT, (D.S., Shifman '20)

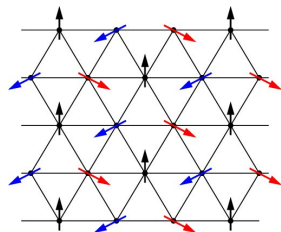
2D triangular antiferromagnet

- Coplanar ground state. Constant spins e_r on sublattices
- Slow variation in spins $S_r(x) = R(x)e_r, \quad R \in SO(3)$.
- Continuum effective theory: take $S_r(x)$ as defined at all x for all r .
- $S_r(x) \cdot S_s(x + a\delta_{r,s}) \sim S_r(x) \cdot \left(S_s(x) + \frac{a^2}{2!} (\delta_{r,s} \cdot \nabla)^2 S_s(x) + \dots \right)$

- Can be rewritten in terms of R field

$$-\partial S_r \cdot \partial S_s \implies \text{Tr} \left[(R^{-1} \partial R)^2 e_s \otimes e_r \right]$$

- This structure $R^{-1} \partial R$ is special

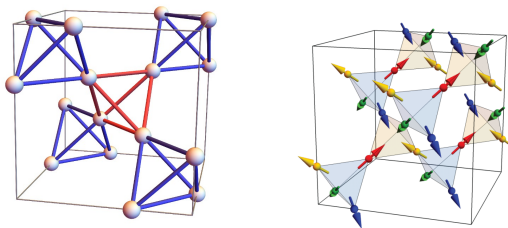


Right translation currents J_μ^k

- $\mathcal{H} \propto \text{Tr} \left[(R^{-1} \partial R)^2 P \right], \quad P \equiv \sum_{s,r} e_s \otimes e_r$
- Describes motion on Lie group: $(R^{-1} \partial_\mu R)_{ij} = 2\epsilon_{ijk} J_\mu^k$
- For typical lattices $P = \text{diag}(a, a, b)$, $a \neq b$ implies squashing!
- $\mathcal{H} \propto (J_\mu^1)^2 + (J_\mu^2)^2 + (1 - \kappa) (J_\mu^3)^2$
- At $\kappa = 1$, gauge invariant with gauge field J^3
- 3D Skyrmion (baryon) charge from Jacobian:

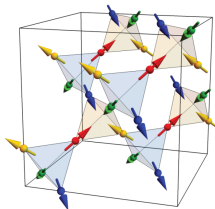
$$Q \propto \int d^3x \epsilon^{\lambda\mu\nu} J_\lambda^1 J_\mu^2 J_\nu^3$$

Pyrochlore lattice



- Rather than triangles, tetrahedra
- Macroscopic degeneracy of classical ground state
- To break this:
 - ▶ Higher order neighbor interactions, especially third
 - ▶ Include quartic terms from the microscopic model $\Rightarrow$ “all-in-all-out”

Toy lattice model



- Triangular or tetrahedral cells are abstracted to a single point
- Rectangular lattice, three spins e_r per site
- Angle between spins fixed by a constraint
- More colinear = more squashed
- Spin e_r only interacts with spin with same r

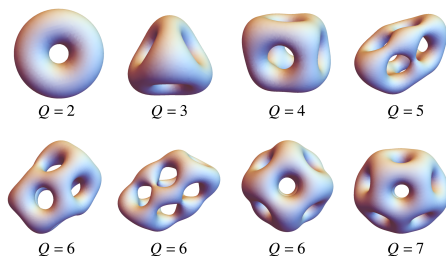
Frustration and stability of defects

- The continuum limit is the squashed sphere model
- Derrick's theorem: We need to keep higher order derivative terms

$$S_r(x + a\delta_r) = S_r(x) + \frac{a^2}{2!} (\delta_r \cdot \nabla)^2 S_r(x) + \frac{a^4}{4!} (\delta_r \cdot \nabla)^4 S_r(x) \dots$$

- But still a sign problem from integration by parts
- Consider ferromagnetic interaction at a and antiferromagnetic at $2a$
- Inversion-symmetric frustrated magnets: Lin, Hayami '16; Sutcliffe '17
- Lifschitz point: Coefficient of 2nd order terms goes to zero

Results of lattice simulation



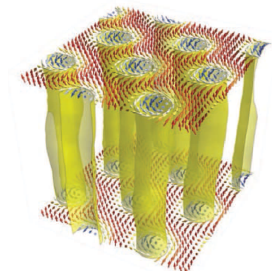
- Very similar to Skyrme model, BPS monopoles
- Rational map ansatz: $U(x) = \cos f(r) + i \sin f(r) \hat{n}(\theta, \phi) \cdot \vec{\sigma}$
- Energy and top charge density of ansatz very close to true solution
- Solutions robust under change of initial conditions
- Switching gears...

Magnetic Skyrmion strings

- In 3D, baby Skyrmions are extended
- 1d core is called 'position curve'
- 2-form to measure baby Skyrmion charge

$$F_{\mu\nu} = \frac{1}{2} \epsilon_{ijk} S^i \partial_\mu S^j \partial_\nu S^k$$

- Can show $dF = 0$ since $S \cdot \partial S = 0$
- Sum of baby Skyrmion charge on a closed surface is zero
- Loops aren't topologically stable?

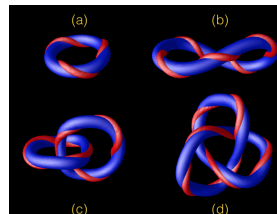
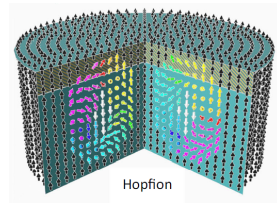


Hopfions

- Just twist the string or tie it in a knot!
- On S^3 , globally defined gauge field $F = dA$
- The Hopf charge is defined as

$$Q = -\frac{1}{8\pi^2} \int d^3x \epsilon^{\lambda\mu\nu} A_\lambda F_{\mu\nu}$$

- Frustrated magnet has Hopfion solutions!
- (Another story: Faddeev-Niemi model and the squashed Skyrme model)

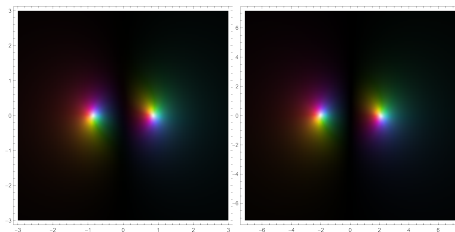


Hopfion charge and baryon charge

- What is the relation between the 3D Skyrmions and Hopfions?
- J^3 is the gauge field. $dJ^3 = F$.
- Another gauge invariant quantity is $J^1 \wedge J^2$
- Flat connection: $F = dJ^3 = 2J^1 \wedge J^2$
- So both charges are the same thing! $A \wedge F \propto J^1 \wedge J^2 \wedge J^3$
- Any field configuration to S^3 can be projected to S^2 with same Q
- Likewise we may (non-uniquely) extend any field configuration to S^2
- The only difference is energy associated to J^3 field

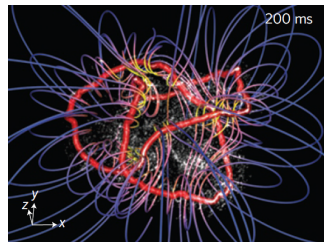
Hedgehog Skyrmions and position curve loops

- How can a spherically symmetric ‘nucleon’ be a loop of string?
- The energy associated to J^1, J^2 concentrated near loop
- J^3 flows through the middle like a dipole field
- The energy and topological charge densities take both into account
- As we ‘squash’ the system, only very subtle change



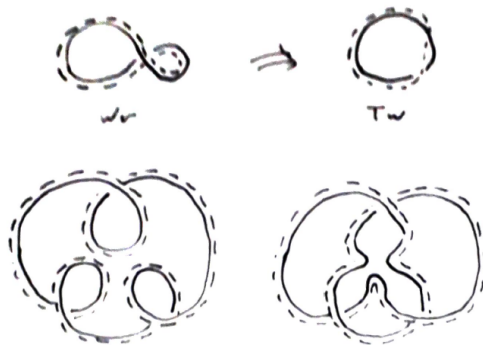
Thin 'baryon strings'

- Ansatz: Unwind hedgehog to form straight string
 - Dipole analogy: Current and magnetic field
 - Can use Biot-Savart law to find J^3 outside core
-
- Fluid analogy: Vorticity and velocity
 - Global vs local strings

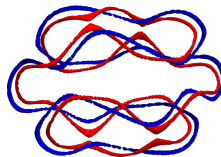
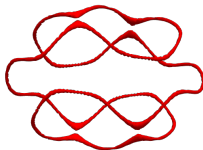
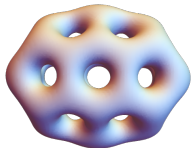
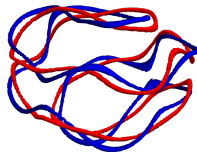
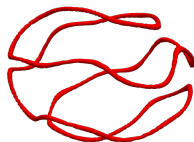
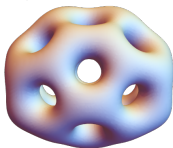
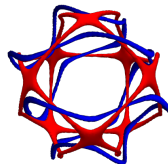
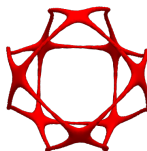
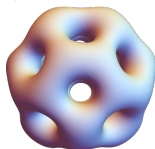


Nuclei as vortex knots?

- Lord Kelvin's old idea
- How can isolated loops form a knot?
- Consider 'ribbon,' $Q = Tw + Wr$

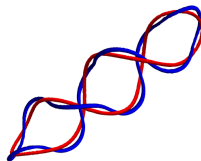
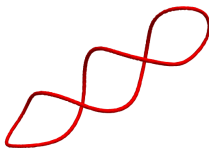
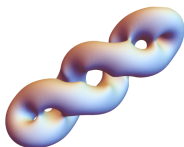


Charge 10 at $\kappa = 0$

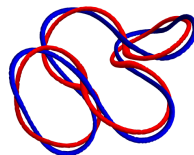
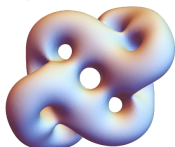
(a) $E/Q = 51.285$ (b) $E/Q = 51.512$ (c) $E/Q = 51.572$ 

Charge 10 at $\kappa = 5/7$

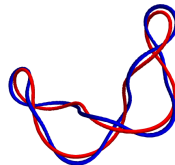
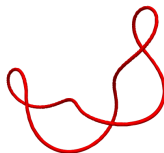
(a) $E/Q = 46.533$



(b) $E/Q = 46.595$

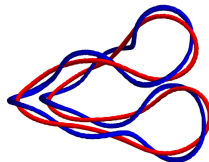
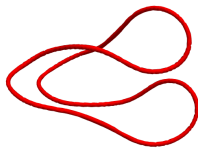
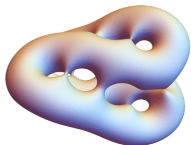


(c) $E/Q = 46.652$

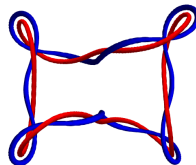
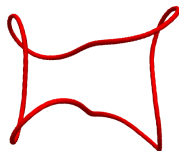
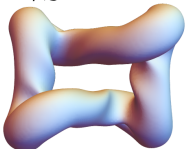


Charge 10 at $\kappa = 5/7$

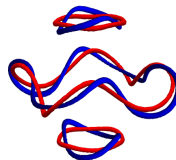
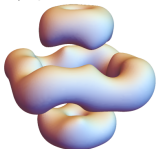
(d) $E/Q = 46.848$



(e) $E/Q = 46.947$

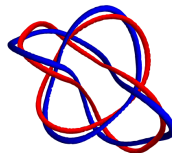
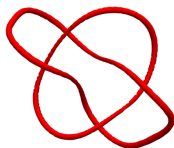
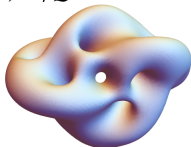


(f) $E/Q = 48.260$

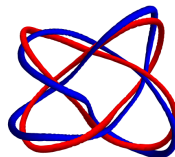
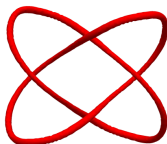
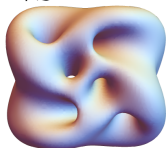


Charge 10 at $\kappa \approx 1$

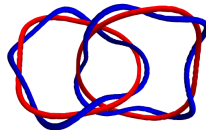
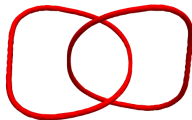
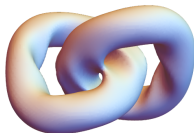
(a) $E/Q = 35.534$



(b) $E/Q = 35.596$

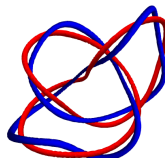
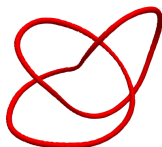
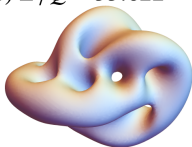


(d) $E/Q = 36.860$



Conclusion

(c) $E/Q = 35.622$

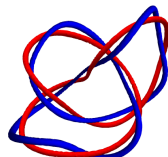
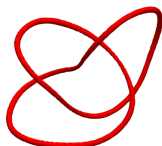
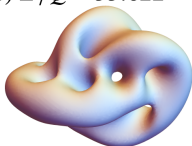


• Future directions

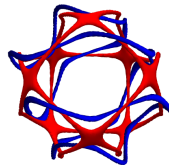
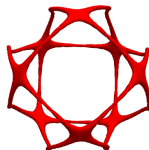
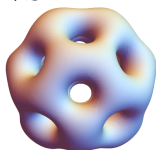
- ▶ Exploring metastable states as κ is varied in more detail
- ▶ 3D Skyrmions in real materials. Pyrochlore lattice?
- ▶ Crossing the Lifschitz point and Skyrmion crystals
- ▶ Any insight into the Skyrme model and QCD?

Thank you!

(c) $E/Q = 35.622$



(c) $E/Q = 51.572$



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